D.K.T.E. Society's Textile and Engineering Institute, Ichalkaranji
(An Autonomous Institute, Affiliated to Shivaji University, Kolhapur)

Accredited with 'A+' Grade by NAAC



Promoting Excellence in
Teaching, Learning & Research

Department of Computer Science and Engineering
(Artificial Intelligence and Machine Learning) 2026-27

THE COMMUNITY ENGAGEMENT PROJECT REPORT ON

"WATER PURIFICATION AWARENESS"

Under the guidance of: Prof. S.S. Demanna

Developed By:

1. Shradha Kumbhar	25UAM030
2. Anuradha Jadhav	25UAM011
3. Ojas Kate	25UAM029
4. Poonam Kokane	25UAM027

CERTIFICATE

This is to certify that,

1. Shradha Kumbhar	25UAM030
2. Anuradha Jadhav	25UAM011
3. Ojas Kate	25UAM029
4. Poonam Kokane	25UAM027

Have successfully completed the project work, entitled

"WATER PURIFICATION AWARENESS"

In partial fulfilment for the award of degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning). This is record of their work carried out during academic year 2026-27.

Prof. S.S. Demanna
[Project Guide]

Prof. Dr. S.K.Shirgave
[Head of Department]

Date:
Place: Ichalkaranji

DECLARATION

We, the undersigned students of S.Y. B.Tech CSE (AIML), declare that the Community Engagement Project report entitled “Water Purification Awareness”, prepared under the guidance of Prof. Rekha Kamble, is our original work.

The information and findings presented in this report are based on field visits, laboratory observations, practical plant-growth activities, discussions, awareness activities, and data collected by us. The content of this report has not been copied from any previously submitted report or other academic work.

Sr. No.	Name	PRN
1	Shradha Kumbhar	25UAM030
2	Anuradha Jadhav	25UAM011
3	Ojas Kate	25UAM029
4	Poonam Kokane	25UAM027

Date:
Place: Ichalkaranji

INDEX

Sr. No.	Contents	Page No.
1	Introduction	1
2	Problem Statement	2
3	Problem Description	3
4	Facility Requirements Related to the Problem	4
5	Field Visit Report	5
6	Technical Solutions for the Problem	6
7	Authority Letter	10
8	Conclusion	11
9	References	12

INTRODUCTION

Water is one of the most essential natural resources required for human life, environmental balance, and community development. Safe and clean water is necessary for drinking, cooking, sanitation, agriculture, and other daily activities. However, water bodies can become polluted due to domestic wastewater, solid waste, chemicals, agricultural runoff, and other contaminants.

Our Community Engagement Project entitled “Water Purification Awareness” focuses on understanding water pollution, exploring a simple plant-based approach for improving water quality, and creating awareness among school students about safe drinking-water practices.

As the first practical activity of the project, our team visited a water body measuring approximately 10 × 10 metres. We studied the site and planned to introduce two selected plant species, Canna indica and Typha angustifolia, as part of a plant-based water treatment approach. These wetland-associated plants can contribute to nutrient uptake and biological treatment processes in suitable systems.

Before introducing the plants into the project water body, our team visited a chemical laboratory and collected/observed plant samples. The plants were maintained under observation for several days to study their growth, survival, and adaptation to the available water conditions.

After the preliminary observation period, the selected plants were transferred to the project water body. During implementation, we identified a practical issue: the water level may increase during rainfall. To address this issue, we developed a simple floating support concept using coconut bark/coconut coir and soil. The floating arrangement provides support to the plants and allows the plant unit to remain near the water surface as the water level changes.

The practical plant-growth and field observation process is planned for approximately two months. During this period, plant growth, survival, water-level changes, and other visible observations are recorded.

The second major part of the project is a school awareness campaign. Our team visits the school and explains the importance of safe drinking water, boiling water when appropriate, proper storage of drinking water, hand hygiene, prevention of water contamination, drainage cleanliness, and water conservation.

Thus, the project combines field study, plant-based environmental work, practical observation, and community awareness. It demonstrates how students can contribute to solving a practical environmental problem while developing teamwork, communication, observation, and social-responsibility skills.

PROBLEM STATEMENT

Water pollution and lack of awareness about safe drinking-water practices are important community and environmental concerns. Water bodies may receive domestic wastewater, solid waste, agricultural runoff, and other pollutants. Poor sanitation and improper drainage can further affect the surrounding environment.

The major financial problems faced by community members include:

- Lack of awareness about safe drinking-water practices.
- Improper disposal of waste near water bodies and drainage systems.
- Possible deterioration of water quality due to pollutants and excess nutrients.
- Need for environmentally friendly approaches for water-body management.
- Changing water levels during rainfall, which can affect fixed plant arrangements.
- Need to study whether selected plants can survive and establish under field conditions.
- Need for awareness among school students about boiling water when appropriate, safe storage, hygiene, and water conservation.

The Water Purification Awareness Project addresses these issues through a practical plant-based activity using Canna indica and Typha angustifolia, along with a school awareness campaign.

PROBLEM DESCRIPTION

The selected project water body has an approximate dimension of 10 × 10 metres. The objective of the practical activity is to explore a natural plant-based approach that can support water-quality improvement while studying plant growth under actual field conditions.

Before placing the plants in the water body, our team visited a chemical laboratory and observed plant samples for several days. This preliminary stage was important for checking plant growth and survival before transferring them to the larger project site.

Canna indica and Typha angustifolia were selected because they are associated with wetland environments and can participate in natural treatment processes through their root zones, plant uptake, and associated microorganisms. The project does not claim that these plants alone make polluted water safe for drinking; laboratory testing is required for any drinking-water safety assessment.

A major practical problem identified during the project was variation in the water level. During rainfall, the water level may increase and fixed plants may become excessively submerged or displaced. Therefore, our team considered a floating support arrangement using coconut bark/coconut coir and soil. The coconut material provides support and buoyancy while the soil provides a medium around the plant roots.

The plant system is planned to be monitored for approximately two months. The team records plant condition, growth, water-level changes, and visible changes in the project site.

In parallel, the school campaign addresses the human side of the water problem. Students are educated about safe drinking water, boiling when appropriate, safe storage, handwashing, prevention of water contamination, and conservation of water.

.FACILITY REQUIREMENTS RELATED TO THE PROBLEM

The following facilities and materials are required for the proposed Water Purification Awareness Project:

1. Project Water Body

An approximately 10 × 10 metre water body for field implementation and observation.

2. Selected Plants

Canna indica and Typha angustifolia for the plant-based practical activity.

3. Chemical Laboratory

A laboratory facility for observing/handling plant samples and understanding preliminary conditions.

4. Plant Observation Facility

Containers or suitable arrangements for keeping plants under observation for several days.

5. Floating Support Material

Coconut bark/coconut coir and soil to support the plants when water levels change.

6. Water Observation

Regular observation of water level, appearance, odour, and other visible field conditions.

7. Measuring and Documentation Tools

Measuring tape, notebook, camera/mobile phone, and other basic field-recording materials.

8. Safety Equipment

Gloves and other basic safety equipment appropriate for field and laboratory work.

9. School Awareness Materials

Posters, charts, presentations, and simple educational material for students.

10. Water Testing

Where available, laboratory water-quality testing before and after the observation period to assess selected parameters.

TECHNICAL SOLUTIONS FOR THE PROBLEM

The following practical and awareness-based solutions can be implemented as part of the Water Purification Awareness Project.

1. Plant-Based Water Treatment

- Description
- Selected wetland-associated plants can be incorporated into suitable water-treatment or constructed-wetland systems. Their roots and surrounding microorganisms can contribute to physical, biological, and nutrient-removal processes.
- Implementation Strategies
- Select healthy Canna indica and Typha angustifolia plants.
- Observe the plants under controlled/preliminary conditions before field transfer.
- Introduce the plants into suitable locations in the project water body.
- Monitor plant growth and survival regularly.
- Where feasible, compare selected water-quality parameters before and after the treatment period.
- Benefits
- Uses a natural biological process.
- Can support nutrient uptake and retention.
- Provides an environmentally friendly supplementary treatment approach.
- Helps students understand ecological water treatment.

2. Preliminary Laboratory Observation

- Description

- Plant samples are observed in a chemical laboratory for several days before field introduction. The purpose is to study plant growth and survival under the available conditions.
- Implementation Strategies
- Collect/obtain selected plant samples.
- Keep samples in suitable containers.
- Observe leaf condition and growth.
- Observe root development where possible.
- Select healthy plants for field implementation.
- Benefits
- Reduces the risk of introducing weak plants.
- Provides initial information about plant adaptation.
- Creates a record of plant growth before field transfer.

3. Floating Plant Support System

- Description
- A simple floating arrangement using coconut bark/coconut coir and soil is used to support the plants. This is intended to help the plant units remain near the water surface when the water level changes.
- Implementation Strategies
- Prepare coconut bark/coir as the floating support.
- Place suitable soil around the plant root zone.
- Secure the plants in the floating unit.
- Place the unit carefully in the water body.
- Observe its stability during water-level changes.
- Benefits
- Uses locally available natural materials.

- Provides plant support.
- Can accommodate changes in water level.
- Reduces dependence on complex infrastructure.

4. Two-Month Monitoring System

- Description
- The project includes approximately two months of observation to study plant growth, survival, and field conditions.
- Record plant height where practical.
- Record number and condition of leaves.
- Observe root development where visible.
- Record water-level changes.
- Record visible changes in water appearance and odour.
- Photograph the plants at regular intervals.

5. Safe Drinking-Water Awareness

- Description
- The school campaign focuses on household and personal practices that help reduce the risk of drinking-water contamination.
- Explain the importance of safe drinking water.
- Explain boiling as a household treatment method when water safety is uncertain.
- Explain safe and covered storage.
- Encourage clean utensils and containers.
- Promote hand hygiene.
- Explain prevention of contamination.

6. Water Conservation and Drainage Awareness

- Description

- The campaign also explains that water quality and sanitation depend on responsible community behaviour.
- Do not throw solid waste into drains or water bodies.
- Keep surroundings clean.
- Avoid unnecessary water wastage.
- Report leaking taps where appropriate.
- Maintain proper drainage practices.

7. School Awareness Campaign

The school campaign provides a direct community-engagement component of the project. Students are informed about safe drinking water, boiling water when appropriate, water storage, handwashing, waste disposal, and conservation.

Activities include:

Interactive discussion with students.

Poster and chart-based awareness.

Question-and-answer session.

Demonstration/explanation of safe water practices.

Discussion of common water-contamination sources.

8. Water Quality Testing

Where facilities are available, water-quality testing can be used to support objective evaluation. Visual appearance alone cannot establish whether water is safe for drinking.

Possible parameters may include physical, chemical, and microbiological indicators selected according to the purpose of the study and available laboratory facilities.

9. Field Documentation

The project should maintain photographs, observation tables, dates, plant-growth records, water-level observations, and notes from the school campaign. Documentation helps evaluate the project over the two-month period.

10. Community Participation

The project encourages cooperation among students, laboratory personnel, municipal/community stakeholders, teachers, and school students. Community participation is essential for maintaining water bodies and promoting safe drinking-water practices.

11. Safety and Environmental Responsibility

The plants and materials used in the project should be handled responsibly. The project should not introduce unknown chemicals or invasive species into the water body. Any water-quality claims should be supported by appropriate testing.

FIELD VISIT REPORT

FIELD VISIT REPORT

Our field activities were conducted as part of the Community Engagement Project titled “Water Purification Awareness.” The activities included a school awareness campaign, a visit to the chemical laboratory, and a visit to the Mahanagarpalika to understand the importance of water purification, sanitation, and public health awareness.

1. Visit to School for Water Purification Awareness

Our team visited a school to create awareness among students about the importance of clean and safe drinking water. We explained the need to drink safe water, maintain personal hygiene, wash hands properly, store drinking water safely, and prevent water contamination. We also discussed the importance of keeping the surroundings clean, maintaining proper drainage, and conserving water.The main objective of this visit was to educate students about safe water practices and encourage them to follow healthy habits in their daily lives.

Image 1: Water Purification Awareness Campaign at School



2. Visit to Chemical Laboratory

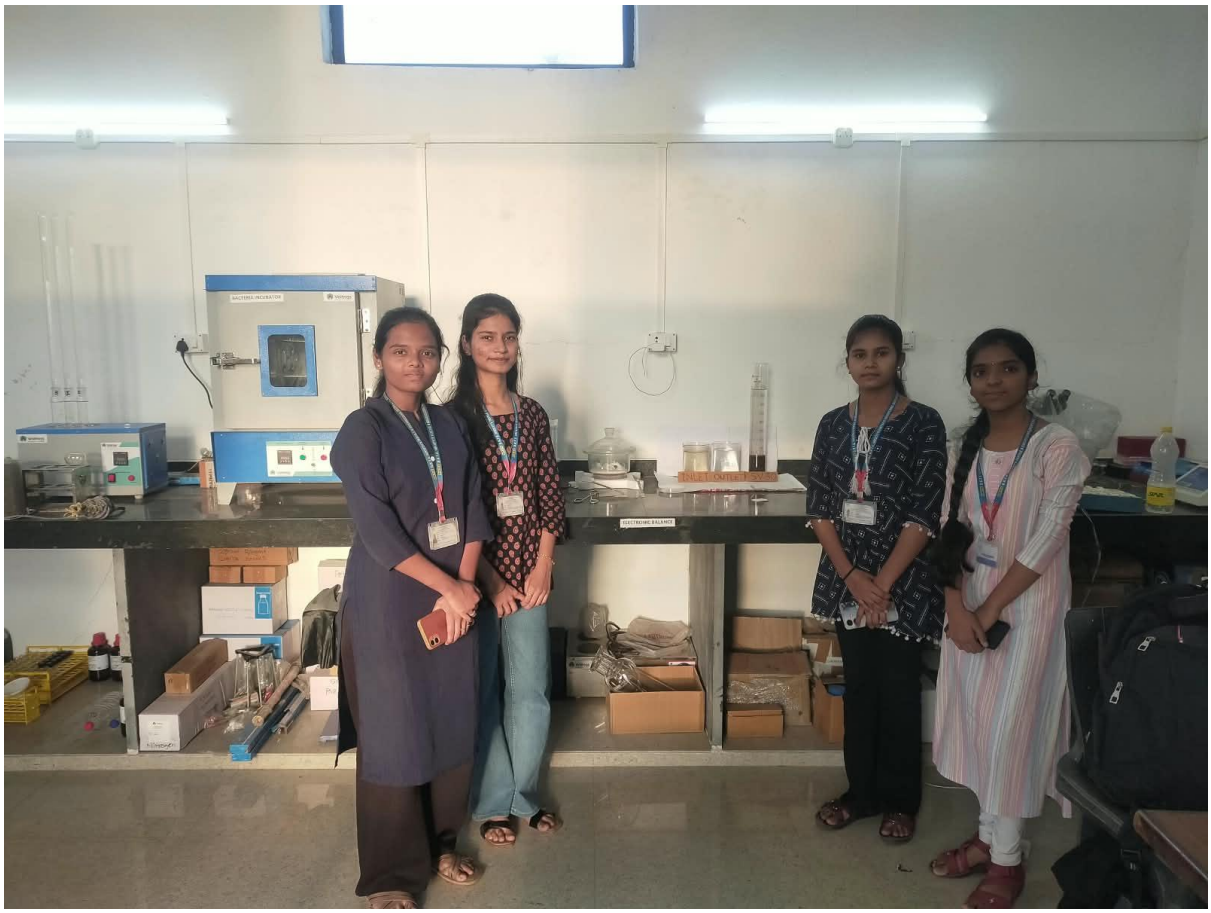
After the school awareness activity, our team visited the chemical laboratory to learn more about water purification and related laboratory materials. During the visit, we observed a water purification awareness chart and laboratory test tubes. The chart helped us understand the topic visually, while the test tubes introduced us to the laboratory equipment used in scientific activities and experiments.

We captured two photographs during this visit: one with the awareness chart and another with the test tubes. This visit helped us connect our project topic with laboratory-based learning.

Image 2: Photograph with the Water Purification Awareness Chart



Image 3: Photograph with Laboratory Test Tubes



3. Visit to Mahanagarpalika

Our team also visited the Mahanagarpalika to understand the importance of municipal services related to water supply, sanitation, drainage, and cleanliness. The visit helped us recognize the role of local municipal authorities in maintaining public hygiene and supporting a clean and healthy environment.

Through this visit, we gained a better understanding of the connection between water management, sanitation, and community health. We also learned about the importance of public participation in keeping water sources and surrounding areas clean.

Image 4: Visit to Mahanagarpalika



4. Overall Learning from the Field Visits

The field visits provided practical learning opportunities related to water purification awareness and community health. The school visit helped us understand the importance of educating students about safe drinking water and hygiene. The chemical laboratory visit provided visual exposure to an awareness chart and laboratory equipment. The Mahanagarpalika visit helped us appreciate the importance of municipal services in water management, sanitation, and cleanliness.

These activities strengthened our understanding of the importance of water awareness and the role that students, educational institutions, and local authorities can play in promoting a cleaner and healthier community.

AUTHORITY LETTER

CONCLUSION

The Water Purification Awareness Community Engagement Project provided our team with an opportunity to combine practical environmental work with community education.

The first part of the project focused on an approximately 10 × 10 metre water body. We explored a plant-based approach using Canna indica and Typha angustifolia. Before introducing the plants into the project water body, we visited a chemical laboratory and observed the selected plant samples for several days to understand their growth and survival.

After the preliminary observation, the plants were transferred to the project water body. During implementation, we identified the challenge of changing water levels. To address this problem, we developed a simple floating support concept using coconut bark/coconut coir and soil. This arrangement is intended to support the plants and keep them near the water surface as the water level changes.

The practical activity is planned for approximately two months. During this period, plant growth, survival, water-level changes, and visible site observations are recorded. Where laboratory facilities are available, water-quality testing can provide objective information about changes in selected water-quality parameters.

The second part of the project involved visiting a school and conducting a Water Purification Awareness Campaign. Students were educated about safe

drinking water, boiling water when appropriate, proper storage, hand hygiene, prevention of water contamination, drainage cleanliness, and water conservation.

The project helped us understand that water management requires both practical environmental approaches and public awareness. Plant-based systems can form part of suitable natural treatment approaches, while awareness and responsible community behaviour are essential for protecting water resources.

Overall, the project helped us develop teamwork, communication, field-observation, problem-solving, environmental-awareness, and community-engagement skills.

Overall, the project promotes:

Safe drinking-water awareness.

Environmentally responsible water management.

Plant-based treatment awareness.

Proper sanitation and drainage practices.

Water conservation.

Community participation.

Environmental responsibility.

In the future, the project can be improved by conducting systematic laboratory testing, maintaining long-term plant-growth records, measuring selected water-quality parameters before and after treatment, and developing a digital awareness platform for the community.

REFERENCES

Books :

1. Kadlec, R. H. and Wallace, S. D. Treatment Wetlands. CRC Press.

2. Metcalf & Eddy. Wastewater Engineering: Treatment and Resource Recovery. McGraw-Hill Education.

3. World Health Organization. Guidelines for Drinking-water Quality. World Health Organization.

Research Papers and Reports :

1. World Health Organization (WHO). Guidelines and resources related to drinking-water quality and household water treatment.

2. UNICEF. Water, Sanitation and Hygiene (WASH) resources.

3. Central Pollution Control Board (CPCB), Government of India. Resources related to water pollution and environmental protection.

4. Bureau of Indian Standards (BIS). IS 10500 – Drinking Water: Specification.

5. Ministry of Jal Shakti, Government of India. Jal Jeevan Mission resources.

Field and Project Sources :

1. Observations collected during the visit to the approximately 10 × 10 metre project water body.

2. Observations collected during the chemical laboratory visit.

3. Plant-growth observations of Canna indica and Typha angustifolia during the approximately two-month project period.

4. Observations of the floating plant-support arrangement using coconut bark/coconut coir and soil.

5. Information and feedback collected during the Water Purification Awareness Campaign at the school.

6. Discussions with laboratory personnel, teachers, students, and community members during the Community Engagement Project.